Weather Quantification for Species Abundance and Diversity using Random Forest, Support Vector Regressor and Neural Network

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Abstract

The purpose of this paper is to record the methodology applied to construct a machine learning model to quantify weather data that indicates potential for meliponine species abundance and species diversity obtained from empirical data. The quantification was accomplished using Random Forest, after comparisons are made with Neural Network and Support Vector Regressor.

Keyword: Random Forest, Neural Network, Support Vector Regressor, meliponine, weather conditions

1. INTRODUCTION

The machine learning approaches have advances that are capable of key information modelling. For quantification of weather conditions that enclosed an area, it is most helpful to create target output variables that can indicate favourable weather conditions for species abundance and species diversity. The documentation is provided to achieve objective that is set to implement the suitable machine learning algorithm for quantifying weather variables that is indicative of species abundance and species diversity. Favourable weather conditions are recognised as important indicators of abundance and diversity in meliponine (or synonymously stingless bees) genera.

In the Diversity Abundance Bipartite (DivAB) network (Jamil, 2026), the foraging hours of meliponines are established as 9-11 AM. Thus, the chosen three weather variables — light intensity, relative humidity, and temperature — which are frequently cited in previous studies (Trianto & Purwanto, 2022; Ramos et al., 2024), are quantified for their potential influence. Light intensity, derived from photosynthetically active radiation (PAR), reflects its role in photosynthesis. The UV index measures ultraviolet exposure. Based on a spatiotemporal meteorology study in Myanmar by Sein et al. (2022), the variables selected such as relative humidity reflects ambient moisture relative to air temperature, while dew point and dry-bulb temperature are direct, location-specific thermal indicators.

The weather abundance score (*Wa*) and weather diversity score (*Wd*) are indicators that summarise dominant quantifiable environmental conditions at each location node corresponding to the observed patterns of abundance and diversity of the bees in the Diversity Abundance Bipartite (DivAB) network (Jamil, 2026). *Wa* and *Wd* are final outputs of the chosen machine learning approach that is directly related to the abundance of the bees at a given location among the 12 bee genera (for *Wa*), while the diversity of the bees is based on genus-level identification among the 35 bee species (for *Wd*).

The making of *Wa* or *Wd* each combines a pre-calculated ratio signifying abundance or diversity, from empirical data with eight variables recorded at a location. Higher positive values of *Wa* and *Wd* indicate a greater expected presence and diversity of bees at a location, considering the weather and location conditions (Jamil, 2026). The methodology for quantification of the weather variables is decided after comparisons of machine learning models created from Random Forest (RF), Neural

Network (NN) and Support Vector Regressor (SVR) that processed 5-variable or 8-variable quantifier, in train–test ratios 70:30 and 80:20. Performance metrics for each quantifier is measured using coefficient of determination (R^2), root mean square error (RMSE) and mean absolute error (MAE). Training loss and validation loss for each quantifier is measured using mean square error (MSE) across train–test ratios.

2. LITERATURE REVIEW

Computation of Numerical Variables using Machine Learning

Available literature in the past ten years had showcased usage of RF, NN and SVR (Probst et al., 2018, Rhodes et al., 2023, Bichri et al., 2024, Thota & Sinha, 2024). Random Forest (RF) is a group of randomised decision trees able to run classification and regression tasks often demonstrating stability and predictive accuracy. Multilayer perceptron (MLP) neural networks enabled the learning of nonlinear mappings through iterative optimisation of weighted connections (Primantara et al., 2024; Moorthy et al., 2024). Support Vector Regressor (SVR) is a method of transformation for input data into higher-dimensional feature spaces via kernels to identify optimal regression hyperplanes within an epsilon-insensitive margin (Karasuyama et al., 2006, Marri et al., 2023, Wang et al. 2025). The summary in Table 1 below.

Table 1. Summary of Several Types of Machine Learning

Type	Feature	Hyperparameter tuning	Referenced Works
Random Forest	Contains the role of geometry in data representation and similarity metrics as mentioned in Rhodes et al. (2023). Suitable for small datasets.	Maximum feature Node sizes Numbers of trees	Probst et al. (2018) applied parameter *max_features*, that provides the highest average improvement in RF model accuracy.
Neural Network	Artificial intelligence utilising neural network architecture similar with biological system of the human dendrites connection.	Epoch Batch size Number of neurons Activation function Number of layers Optimiser type Threshold values	Bichri et al. (2024) selected batch size of 32 and epoch of 60 in image processing application of neural network.
Support Vector Regressor	Requires tuning strategy due to wide range of test values for epsilon (ε) values and regularisation parameter or cost values (C).	Kernel types Epsilon Cost values	Thota & Sinha (2024) applied a grid search with five-fold cross-validation (CV = 5).

Utilising standard performance metrics

Coefficient of Determination (R^2) measures the proportion of variance explained by the model. Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) both calculate the estimation error showing on average how many units does the model's generated score deviates from the pre-calculated reference score. Training and validation losses are measured by Mean Square Error (MSE).

3. METHODOLOGY

The study uses MARDI data on the abundance and diversity of stingless bees in Malaysia (Jaapar et al., 2016; Jamil et al., 2024; Jamil 2026) and the hourly satellite data of the habitat are from the Prediction of Worldwide Energy Resource (POWER) Project's Hourly v2.2.16 version. The stingless bee observation data is further processed into the following ratios defined as in equations which are diversity in genera abundance, *M_gdv*, in equation (1), and bee abundance in a species, *S_cpg,* in equation (2). *S_cpg* and *M_gdv* are computed as numerical variable, which are used as the target variables in the quantification employing machine learning approaches for producing weather scores for abundance (*Wa*) and diversity (*Wd*), respectively.

$$M_gdv = \sum_{n=1}^{12} M_{n_gdv_L} \quad \text{(Equation 1)}$$

Where $M_{n_gdv_L} = \frac{\text{count of species occurrences within a genus at a location}}{\text{total count of species occurrences within the same genus across all 12 locations}}$

$$S_cpg = \frac{number\ of\ individual\ bees\ species\ at\ a\ specific\ location}{total\ number\ of\ individual\ bees\ of\ the\ same\ species\ across\ all\ 12\ locations} \quad \text{(Equation 2)}$$

Random forest (RF) regression model is implemented using the scikit-learn Python in a Google Colab Jupyter Notebook environment. Probst et al. (2018) mentioned the tuning parameter called maximum features (*max_features*) which is typically set to the square root of the total number of features, that provides the highest average improvement in RF model accuracy. Parameter tuning for RF in this study involved selecting from five node sizes (*NS*) (1, 2, 3, 4, 5) and three numbers of trees (10, 50, 100), adapting the work of Rhodes et al. (2023) about testing combinations of node sizes (1, 5, 10, 20, 50) and numbers of trees (5, 10, 50, 100, 250), which emphasises the role of geometry in data representation and similarity metrics in random forest that preserves accuracy within RF models. Five different *NS* choices (1 to 5) and three tree counts (10, 50, 100) are compared for generating *Wa*, weather score abundance models and *Wd*, weather diversity models using RF. The tuning parameters are summarised in Table 2. Models M1 (8 variables) and M2 (5 variables) are trained using 80:20 and 70:30 data splits as shown in Table 3.

Table 2. Random Forest Tuning Parameters

Parameter	Description	Tested values
Maximum feature	Number of features considered when splitting a node	M1, M2
Node size (leaf)	Minimum number of samples required to split an internal node	1, 2, 3, 4, 5
Number of trees	Total number of decision trees grown in the forest	10, 50, 100

Table 3. Variables Grouped for Testing

M1	M2
1. Elevation, *El* 2. Latitude 3. Longitude 4. Light intensity, *LX* 5. Ultraviolet ray, *Index UV* 6. Average air on dry bulb, *T2M* 7. Dew/frost point temperature, *T2MDEW* 8. Relative humidity, *RH2M*	1. Light intensity, *LX* 2. Ultraviolet ray, *Index UV* 3. Average air on dry bulb, *T2M* 4. Dew/frost point temperature, *T2MDEW* 5. Relative humidity, *RH2M*

For comparison, Neural Network (NN) and Support Vector Regressor (SVR) are also trained in the same environment using scikit-learn. The tuning parameters are tabulated in Table 4 for NN and Table 5 for SVR. The hyperparameter tuning for NN quantifier includes, epoch, batch size, number of neurons, activation function, number of layers, optimiser type, and threshold values. The activation functions used for two different layers are 2-layer ('ReLU', 'ReLU', 'linear') and 3-layer ('ReLU', 'ReLU', 'ReLU', 'linear'). The three optimisers considered are 'Adaptive Moment' or 'Adam', 'Root Mean Square Propagation' or 'RMSprop' and 'Stochastic Gradient Descent' or 'SGD'.

The feedforward multi-layer perceptron (MLP) algorithm of NN is implemented in this study, with these three optimisers, the number of neurons set to 5, 10, or 15, and threshold values of 0.001, 0.05, and 0.5. Hence, performance metrics for every NN quantifier with each combination of hyperparameter are determined. One example of the hyperparameters for an NN quantifier is a three-layer network with 5 neurons per layer (5-5-5), a threshold value of 0.001 and using the optimiser, 'Adam'. The combination of hyperparameters is first used to determine the number of layers with the highest performance. Next, the hyperparameter tuning is used to determine the batch size of 3 or 32, and epochs of 60 or 100 epochs at the same threshold values, using the same optimiser types. The batch size of 32 and epoch of 60 are selected for comparison based on the work of Bichri et al. (2024).

Table 4. Tuning Parameters for Neural Network

Parameter	Description	Tested values
Epoch	Number of complete passes through the training dataset	60, 100
Batch size	Number of samples processed before the model is updated	3, 32
Number of neurons	Units in each hidden layer controlling model capacity	5, 10, 15
Activation function	Non-linear transformation applied at each layer; defines output behaviour	ReLU (Input), Linear (Output)
Number of layers	Depth of the network or number of hidden layers	2, 3
Optimiser type	Algorithm used to update weights during training	Adam, RMSprop, SGD
Threshold values	Cut-off values used for evaluation or decision criteria	0.001, 0.05, and 0.5

The hyperparameter tuning for SVR quantifier was developed through a grid search with five-fold cross-validation (CV = 5), following the approach of Thota & Sinha (2024). The cross-validation is executed on the testing of different epsilon (ε) values (0.01, 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5) and varying regularisation parameter or cost values (C) (0.01, 0.1, 1, 10, 100, and 1000). The feature range of exponents for the grid search follows the work of Velasco et al. (2022) and Hoque and Aljamaan (2021). In addition, three kernel types, 'Linear', 'Polynomial' and 'Radial Basis Function' or 'Radial', are evaluated (Ding et al., 2021). In comparison, Hoque and Aljamaan (2021) employed ε values of 0.001, 0.01, 0.1, 1.0 and 10. This present tuning strategy was used with reference to the work of Karal (2024) to examine the variation in model performance with respect to the cost parameters. Similar to Karal (2024), other studies involving SVR — sometimes alongside Random Forest as a separate model (Abbasi et al., 2021; Sakaa et al., 2022) — have explored the relationship between epsilon and cost, while the effects of kernel type have been emphasized by Singh et al. (2024). The tuning parameters for SVR are summarised in Table 5.

Table 5. Tuning Parameters for Support Vector Regressor

Parameter	Description	Tested values
Epsilon values (ε)	Defines the margin of tolerance where no penalty is given to prediction errors within this range	0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5
Regularization parameter or cost values (C)	Controls the trade-off between model complexity and error tolerance; higher values allow less error but risk overfitting	0.01, 0.1, 1, 10, 100, 1000
Kernel types	Functions that transform input data into higher-dimensional space to capture non-linear relationships	Linear, Polynomial, Radial Basis Function (RBF)

In detail, the performance metrics are chosen according to Chicco et al. (2021) stating R^2, RMSE, MAE and MSE are significant predictors across disciplines (Ozili, 2023). Neither paper provided the foundational derivations for these standard metrics. Ozili (2023) discussed the practical application of R^2 in social science stating that low R^2 values (at least 0.1) can be acceptable if predictors are statistically significant. While R^2 measures predictive consistency, MSE, RMSE, and MAE calculate deviation magnitudes. Lower values for error-based metrics (MSE, RMSE, MAE) are preferred as they indicate higher model accuracy. Higher R^2 values are generally preferred, representing a better model fit. The following equations define the standard metrics for assessing model performance, where y_i represents the actual value, $\bar{y}$ represents the mean of the observed data, $\hat{y}_i$ represents the predicted value, and n is the number of observations:

In Equation 3, R^2 measures how much of the variation in the dependent variable is explained by the model. Higher values indicate better performance; 1 is the ideal value, while negative values indicate poor predictive performance. RMSE measures the average prediction error while assigning greater weight to large errors and outliers. Lower values indicate better performance, with 0 being ideal.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad \text{(Equation 3)}$$

In Equation 4, MAE measures the average absolute difference between observed and predicted values. It is less sensitive to outliers than RMSE. Lower values indicate better performance, with 0 being ideal.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|(y_i - \hat{y}_i)| \quad \text{(Equation 4)}$$

In Equation 5, MSE measures the average squared difference between observed and predicted values. It is sensitive to large errors and outliers; lower values indicate better performance, with 0 being ideal.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad \text{(Equation 5)}$$

In Equation 6, RMSE is the square root of MSE, it measures the square root of average squared difference between observed and predicted values. It is sensitive to large errors and outliers; lower values indicate better performance, with 0 being ideal.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad \text{(Equation 6)}$$

The overall methodology is summarised in Figure 1.

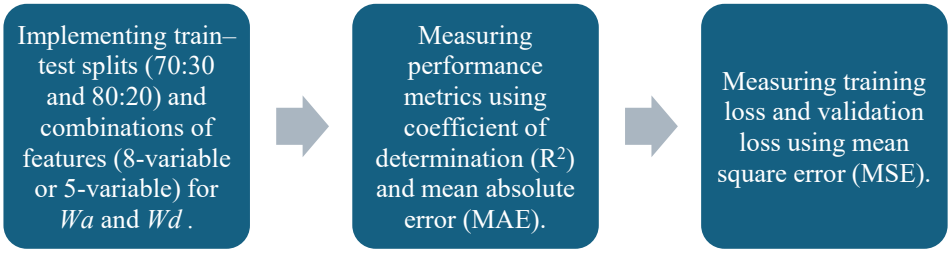


Figure 1. Methodology Flowchart

4. RESULTS AND ANALYSIS OF FINDINGS

The quantification resulted in ML models or quantifiers used for the computation of weather score for the abundance (*Wa*) and weather score for the diversity (*Wd*). Performance metrics of the quantifiers from computation of R^2, RMSE and MAE are presented. The chosen machine learning is Random Forest (RF) classifier with tuning customised for **M1 80:20**. Other results from comparing the train–test split effects in Neural Network (NN), Support Vector Regressor (SVR) and Random Forest (RF) are presented at the end of the section.

Table 6. Random Forest using fixed Node Size '3'

ntree	Group	Ratio	Target	Leaf	R^2	RMSE	MAE
10	**M1**	**70:30**	*Wa*	3	0.883	0.080	0.058
50	**M1**	**70:30**	*Wa*	3	0.859	0.088	0.069
100	**M1**	**70:30**	*Wa*	3	0.891	0.077	0.060
10	**M2**	**70:30**	*Wa*	3	0.508	0.164	0.131
50	**M2**	**70:30**	*Wa*	3	0.545	0.158	0.125
100	**M2**	**70:30**	*Wa*	3	0.565	0.154	0.121
10	**M1**	**80:20**	*Wa*	3	**0.724**	**0.066**	**0.041**
50	**M1**	**80:20**	*Wa*	3	0.839	0.051	0.037
100	**M1**	**80:20**	*Wa*	3	0.830	0.052	0.037
10	**M2**	**80:20**	*Wa*	3	-0.593	0.160	0.119
50	**M2**	**80:20**	*Wa*	3	-0.334	0.146	0.103
100	**M2**	**80:20**	*Wa*	3	-0.225	0.140	0.100
10	**M1**	**70:30**	*Wd*	3	0.902	0.072	0.054
50	**M1**	**70:30**	*Wd*	3	0.852	0.088	0.070
100	**M1**	**70:30**	*Wd*	3	0.831	0.094	0.075
10	**M2**	**70:30**	*Wd*	3	0.438	0.172	0.132
50	**M2**	**70:30**	*Wd*	3	0.443	0.172	0.135
100	**M2**	**70:30**	*Wd*	3	0.459	0.169	0.131
10	**M1**	**80:20**	*Wd*	3	**0.722**	**0.088**	**0.064**

50	**M1**	**80:20**	*Wd*	3	0.719	0.088	0.068
100	**M1**	**80:20**	*Wd*	3	0.775	0.079	0.061
10	**M2**	**80:20**	*Wd*	3	-0.229	0.185	0.127
50	**M2**	**80:20**	*Wd*	3	-0.056	0.171	0.128
100	**M2**	**80:20**	*Wd*	3	-0.002	0.167	0.124

In the Table 6, the mentioned classifier setting was decided: Number of Trees '10' and Node Size '3' (*NS* = 3, *ntree* = 10). The performance metrics are presented for comparisons by number of trees and number of variables, eight-variable (M1) or five-variable (M2). The effects of omitting latitude, longitude and elevation are in M2 types, which are less coefficient of determination (R^2) obtained, and relatively higher error metrics measured (RMSE and MSE). The chosen setting M1 80:20 for *Wa* and *Wd* have approximately similar R^2 (0.724 and 0.722) while error metrics are relatively low, with RMSE computed 0.066 and 0.088 respectively, and MAE 0.041 and 0.064. The decided Node Size is '3' based on Figure 2 and Number of Trees is '10' based on Figures 3 and 4.

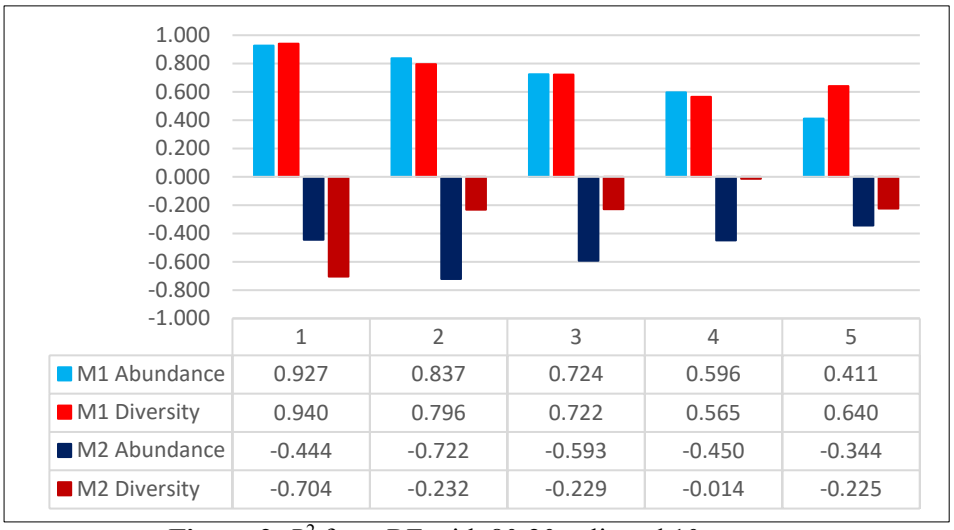


Figure 2. R^2 from RF with 80:20 split and 10 trees

Figure 2 shows M1 *Wa* (Abundance) and *Wd* (Diversity) models, R^2 declined as NS increased from '1' to '5,' with the highest values at *NS* = 1 followed by *NS* = 2, and *NS* = 3 ranked third. For *Wa*, the node sizes when increased from '1' to '3', the resulting R^2 values are 0.927, 0.837, and 0.724 for 10, 50, and 100 trees respectively. For *Wd*, the node sizes when increased from '1' to '3', the resulting R^2 values are 0.940, 0.796, and 0.722 for '10', '50', and '100' trees respectively. The coefficients computed for *Wa* and *Wd* show that they are approximate equals reported in exact measures, around ± 0.0041 at maximum difference of each other. The lower R^2 in *NS* = 4 (0.596, 0.565) and *NS* = 5 (0.411, 0.640) excluded them from further comparative analysis.

Selecting *NS* = 3 which is a midpoint between 1 and 5 is suitable to strike a balance between underfitting and overfitting in a small sample size of 12 rows with 5 or 8 features. This selection enables the model to learn the patterns in the features to acquire meaningful data structures, an approach supported by the theoretical frameworks of Breiman (2004) and Biau (2012) on random forest regression. The research emphasised that by allowing more branching in the trees, the setting becomes highly effective, automatically smooths out errors and this prevents overfitting by averaging the predictions of all trees together (Ishwaran, 2015). In other words, when *NS* = 3 for a small dataset (12×5 or 12×8), the base estimators in the trees are enabled to grow sufficiently to finally average their answers together. The larger node sizes (4 or 5) would force the trees to become too shallow, causing underfitting

because the model could not learn the underlying patterns sufficiently within the 5 or 8 features. Figures 3 and 4 shall demonstrate the trees effect with Node Size fixed at '3' for *Wa* and *Wd.*

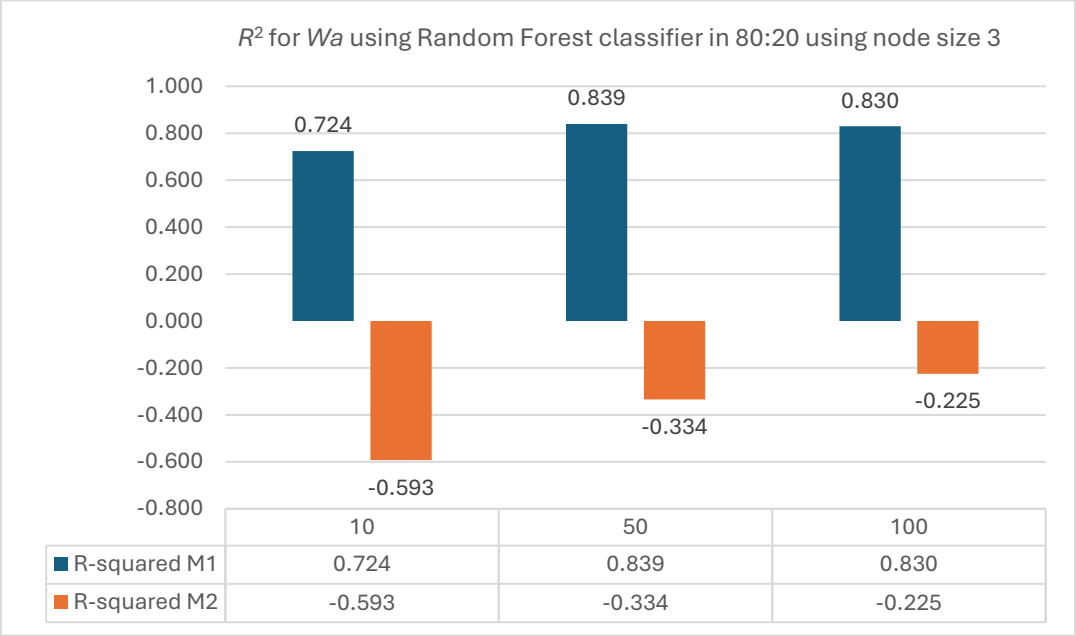


Figure 3. R^2 from chosen classifier settings of trees generated from *Wa*

Figure 3 shows the graph for '10', '50' and '100' trees to M1 and M2 that correspond to the computed R^2 in node size '3'. All M2 has negative results. The chosen was M1 with '10' trees for *Wa* with R^2 0.724. Thus, the 8 variables in M1 explain 72% of the variance in the target variable, *Wa*.

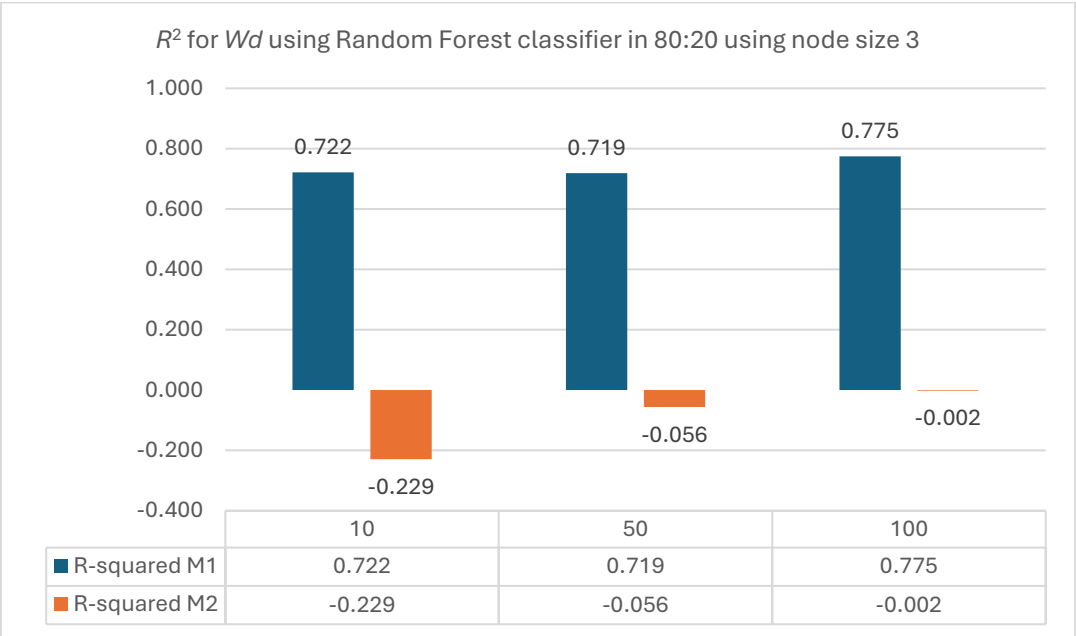
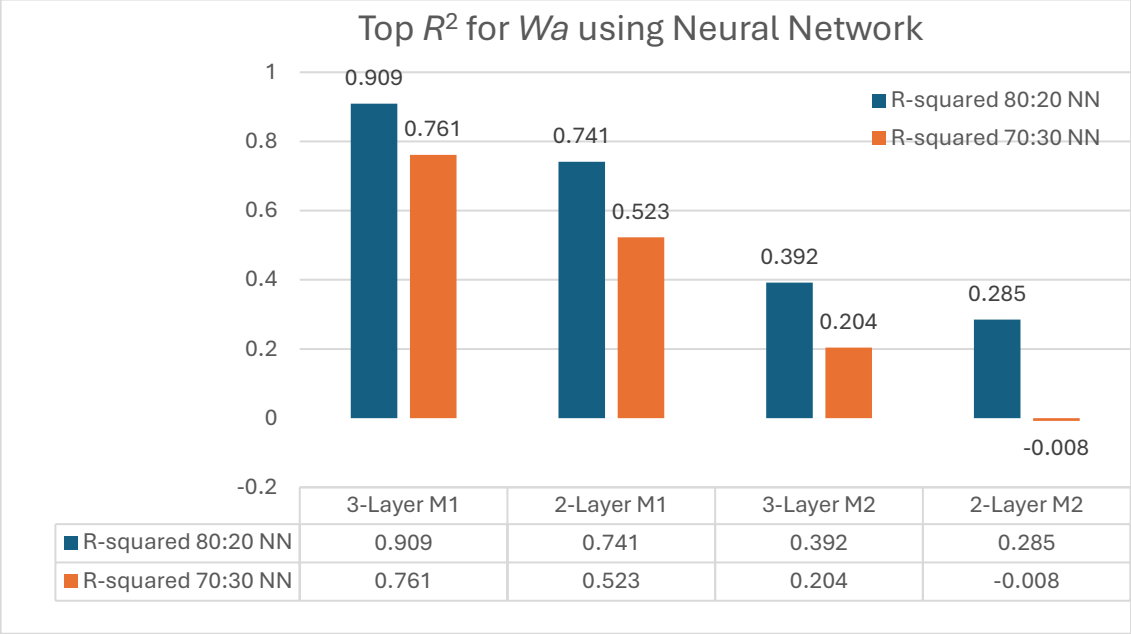


Figure 4. R^2 from chosen classifier settings of trees generated from *Wd*

Based on Figure 4, the same pattern of R^2 is observed for generating *Wd.* The similarity is where the negative values occurred in all M2 models which have lesser number of variables as shown in Figure 3. The chosen was M1 with '10' trees for *Wd*. Thus, the R^2 also explained 72% of variance of the

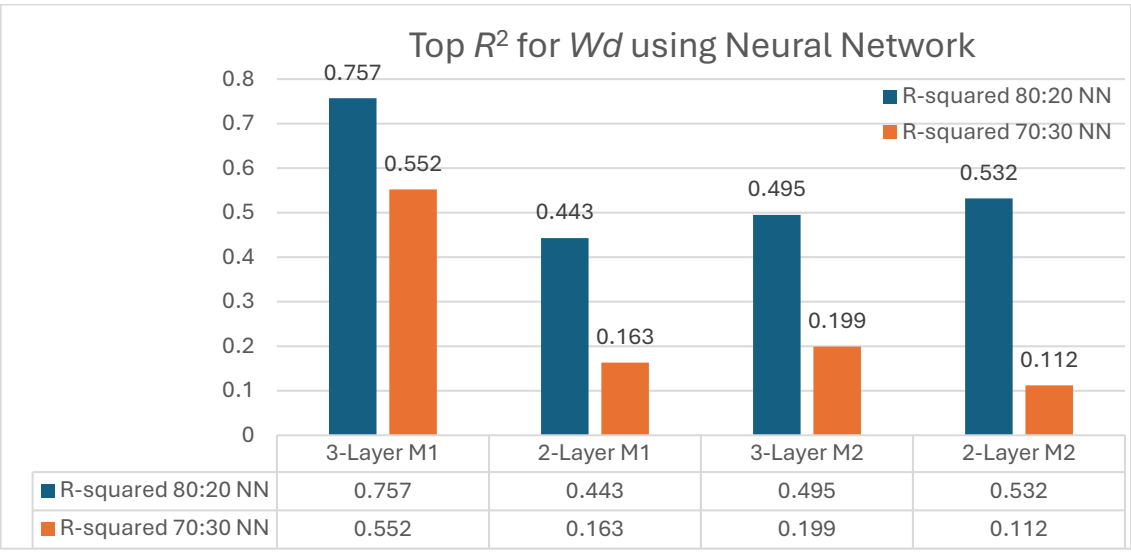
variables causing *Wd*. Therefore, the greater number of trees (100) have been seen with higher R^2 as seen in Figure 3 for *Wa* (0.830) and for *Wd* (0.775), while lesser number of trees (10) have lower R^2 as seen in *Wa* (0.724) and *Wd* (0.722). This finding is consistent with the mention of more trees allowed to branch, the highly effective is the classifier. However, '10' trees is chosen instead of '50' or '100' to avoid overfitting risk.

Thus, the results from another type of quantifier are presented. Top R^2 are presented from the utilisation of 2-layer or 3-layer in Figure 5 and Figure 6. Table 7 shows in detail the optimiser, threshold and layer structure with the error metrics where there is no SGD optimiser in top R^2 list due to lower gain in R^2. However, the best quantifier is 3-layer type with '15' neurons which are further tuned and presented in Table 8 and Figure 7, showing all optimisers compared including SGD.



	3-Layer M1	2-Layer M1	3-Layer M2	2-Layer M2
R-squared 80:20 NN	0.909	0.741	0.392	0.285
R-squared 70:30 NN	0.761	0.523	0.204	-0.008

Figure 5. Top R^2 from layer effects to generate *Wa*



	3-Layer M1	2-Layer M1	3-Layer M2	2-Layer M2
R-squared 80:20 NN	0.757	0.443	0.495	0.532
R-squared 70:30 NN	0.552	0.163	0.199	0.112

Figure 6. Top R^2 from layer effects to generate *Wd*

Table 7. Neural Network Top R^2 Compared to Fixed Batch Size

Layer	Group	Ratio	Batch size	Target	R^2	RMSE	MAE	Optimiser	Threshold	Layer structure
3-Layer	**M1**	**80:20**	3	*Wa*	0.909	0.08	0.062	Adam	0.5	15,15,15
2-Layer	**M1**	**80:20**	3	*Wa*	0.741	0.135	0.115	Adam	0.5	15,15
3-Layer	**M2**	**80:20**	3	*Wa*	0.392	0.207	0.177	RMSprop	0.05	15,15,15
2-Layer	**M2**	**80:20**	3	*Wa*	0.285	0.225	0.184	Adam	0.5	10,10
3-Layer	**M1**	**70:30**	3	*Wa*	0.761	0.133	0.102	RMSprop	0.5	15,15,15
2-Layer	**M1**	**70:30**	3	*Wa*	0.523	0.187	0.153	Adam	0.5	15,15
3-Layer	**M2**	**70:30**	3	*Wa*	0.204	0.242	0.22	Adam	0.05	10,10,10
2-Layer	**M2**	**70:30**	3	*Wa*	-0.008	0.272	0.228	Adam	0.5	10,10
3-Layer	**M1**	**80:20**	3	*Wd*	0.757	0.126	0.108	Adam	0.5	15,15,15
2-Layer	**M1**	**80:20**	3	*Wd*	0.443	0.191	0.155	RMSprop	0.001	10,10
3-Layer	**M2**	**80:20**	3	*Wd*	0.495	0.181	0.141	RMSprop	0.5	10,10,10
2-Layer	**M2**	**80:20**	3	*Wd*	0.532	0.175	0.133	Adam	0.001	15,15
3-Layer	**M1**	**70:30**	3	*Wd*	0.552	0.163	0.121	Adam	0.05	10,10,10
2-Layer	**M1**	**70:30**	3	*Wd*	0.163	0.222	0.187	Adam	0.001	10,10
3-Layer	**M2**	**70:30**	3	*Wd*	0.199	0.217	0.179	RMSprop	0.05	10,10,10
2-Layer	**M2**	**70:30**	3	*Wd*	0.112	0.229	0.184	Adam	0.001	15,15

Based on Table 7, optimiser 'Adam' appears with threshold '0.5' most often in various layer structures and ratio. In much lower threshold (0.001) 'Adam' optimiser shows low R^2 (0.532, 0.163, 0.112). 'RMSprop' shows high R^2 (0.761) in 3-layer for M1 70:30 targeting *Wa* but a low R^2 (0.443) in 2-layer for M1 80:20 targeting *Wd*. Table 8 presents performance metrics for the fixed settings made of batch size '3', three layers of '15' neurons, epoch '100', and threshold '0.5' for all optimisers targeting the output variable *Wa* or *Wd*, all in the determined quantifier type M1 80:20 using Neural Network.

Table 8. Neural Network M1-type with 3 Layers of '15' Neurons at Epoch '100', batch size '3', and $t = 0.5$ for 80:20

Optimiser	Group	Ratio	Target	R^2	RMSE	MAE
Adam	**M1**	**80:20**	*Wd*	0.810	0.111	0.085
RMSprop	**M1**	**80:20**	*Wd*	0.288	0.215	0.184
SGD	**M1**	**80:20**	*Wd*	0.129	0.238	0.194
Adam	**M1**	**80:20**	*Wa*	0.770	0.120	0.094
RMSprop	**M1**	**80:20**	*Wa*	0.572	0.174	0.132
SGD	**M1**	**80:20**	*Wa*	0.392	0.207	0.175

Comparing the overall performance, optimiser 'Adam' proved to be the superior among the three compared, resulting in the highest R^2 scores twice (0.810 and 0.770), with the lowest RMSE (0.111 and 0.120) and MAE (0.085 and 0.094), in both targeting of *Wd* and *Wa*. In contrast, 'SGD' showed it worked in much reduced performance with R^2 values measured (0.129 and 0.392) in targeting both *Wd* and *Wa* respectively. The demonstration of lowest R^2 values appeared twice as mentioned with the highest error in both RMSE and MAE respectively, in *Wd* (0.238 and 0.194) and *Wa* (0.207 and 0.175). Thus, the lowest RMSE and MAE is generated by 'Adam' optimiser. The highest RMSE and MAE is generated by 'SGD' optimiser. For visualisation, Table 8 is presented in Figure 7 showing R^2. Further effects in tuning of epoch, batch size and thresholds on training loss and validation loss are shown in the Table 9.

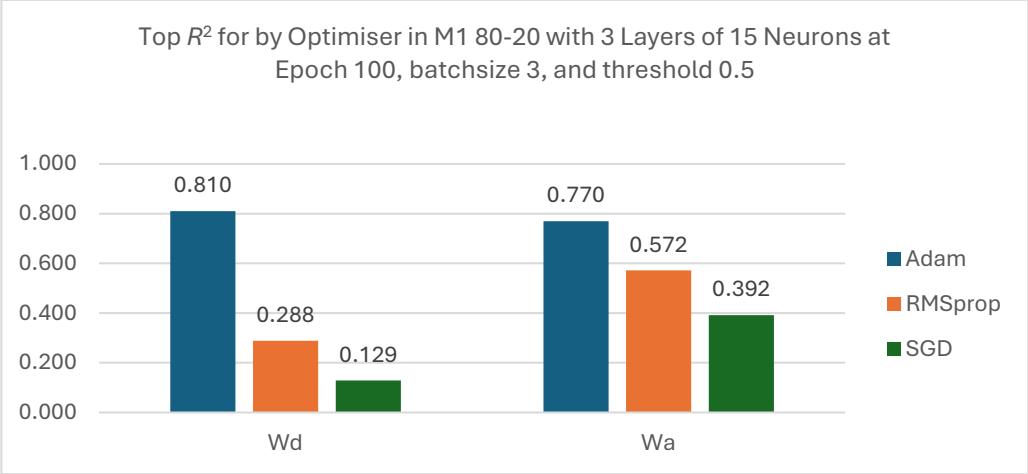


Figure 7. Top R^2 from different optimisers

Table 9. Effects of batch sizes and epoch in M1 80:20 with 3 layers of 15 neurons in training loss, validation loss and the resulting threshold-optimiser combinations

Batch size	Epoch	Target	Threshold	Optimiser	Training Loss (MSE)	Validation Loss (MSE)
32	60	*Wd*	0.001	SGD	0.058	0.079
32	60	*Wd*	0.001	Adam	0.055	0.073
32	100	*Wd*	0.5	SGD	0.060	0.072
32	60	*Wd*	0.5	Adam	0.060	0.072
32	60	*Wa*	0.001	SGD	0.065	0.071
32	60	*Wa*	0.5	RMSprop	0.056	0.069
3	60	*Wd*	0.001	RMSprop	0.048	0.068
32	100	*Wa*	0.5	Adam	0.058	0.067
32	100	*Wd*	0.001	RMSprop	0.058	0.067
3	100	*Wd*	0.001	SGD	0.051	0.050
3	100	*Wa*	0.5	SGD	0.041	0.050
3	100	*Wa*	0.001	SGD	0.047	0.048
3	60	*Wa*	0.5	RMSprop	0.031	0.039
3	100	*Wd*	0.5	RMSprop	0.037	0.037
3	100	*Wa*	0.001	RMSprop	0.028	0.031
3	100	*Wd*	0.5	Adam	0.017	0.020
3	100	*Wd*	0.001	Adam	0.030	0.019
3	100	*Wa*	0.5	Adam	0.009	0.013

Consequently, the decided quantifier type (M1 80:20) is further tuned in fixed number of batch and fixed epoch, shown in Table 9 measured in training loss and validation loss. Thus, it is found that high training loss and validation loss are found in larger batch (32) while smaller batch (3) have lower measurements of the MSE. The results are sorted on descending order of validation loss showing outcomes from comparison of epoch (‘60’ vs ‘100’) and batch (‘32’ vs ‘3’) when further tuning is run on all three optimisers on fixed parameters: 3-layer (15,15,15) at thresholds ‘0.5’ and ‘0.001’. Bichri et al. (2024) selected batch size ’32’, epoch ’60’, and threshold value of ‘0.01’, applied ‘Adam’ optimiser and concluded that the train–test split ratio has a significant impact on performance in custom datasets

for deep learning models. For the comparison of quantifier type using neural network, this study found that 'Adam' optimiser, batch size '3', epoch '100', threshold '0.5' and layer structure (15,15,15) in M1 80:20 from Table 8 data suggesting the setting is suitable for targeting *Wd* (0.810 R^2, 0.111 RMSE, 0.085 MAE) and *Wa* (0.770 R^2, 0.120 RMSE, MAE 0.094).

Table 10. Support Vector Regressor in fixed epsilon with resulting cost and performance metrics

Kernel	Group	Ratio	Target	Epsilon	Resulting Cost	R^2	RMSE	MAE
RADIAL	**M1**	**80:20**	*Wa*	0.01	100	0.942	0.063	0.048
RADIAL	**M1**	**70:30**	*Wd*	0.01	10	0.767	0.107	0.076
RADIAL	**M1**	**80:20**	*Wd*	0.01	1000	0.804	0.108	0.077
RADIAL	**M2**	**80:20**	*Wd*	0.01	10	0.773	0.116	0.085
RADIAL	**M1**	**70:30**	*Wa*	0.01	1	0.657	0.130	0.096
POLYNOMIAL	**M1**	**80:20**	*Wa*	0.01	100	0.699	0.143	0.103
RADIAL	**M2**	**70:30**	*Wa*	0.01	1	0.367	0.177	0.104
RADIAL	**M2**	**70:30**	*Wd*	0.01	1	0.367	0.177	0.104
POLYNOMIAL	**M2**	**80:20**	*Wd*	0.01	1	0.606	0.153	0.109
POLYNOMIAL	**M1**	**80:20**	*Wd*	0.01	100	0.612	0.151	0.110
LINEAR	**M2**	**70:30**	*Wa*	0.01	1000	0.448	0.166	0.111
LINEAR	**M2**	**70:30**	*Wd*	0.01	1000	0.448	0.166	0.111
POLYNOMIAL	**M1**	**70:30**	*Wa*	0.01	1	0.592	0.142	0.113
POLYNOMIAL	**M1**	**70:30**	*Wd*	0.01	1	0.592	0.142	0.113
LINEAR	**M2**	**80:20**	*Wd*	0.01	100	0.541	0.165	0.116
LINEAR	**M1**	**80:20**	*Wd*	0.01	1000	0.499	0.172	0.119
LINEAR	**M1**	**70:30**	*Wa*	0.01	10	0.432	0.168	0.132
LINEAR	**M1**	**70:30**	*Wd*	0.01	10	0.432	0.168	0.132
POLYNOMIAL	**M2**	**70:30**	*Wa*	0.01	10	0.219	0.197	0.135
POLYNOMIAL	**M2**	**70:30**	*Wd*	0.01	10	0.219	0.197	0.135
POLYNOMIAL	**M2**	**80:20**	*Wa*	0.01	1	0.530	0.179	0.158
RADIAL	**M2**	**80:20**	*Wa*	0.01	10	0.398	0.202	0.162
LINEAR	**M1**	**80:20**	*Wa*	0.01	100	0.448	0.194	0.164
LINEAR	**M2**	**80:20**	*Wa*	0.01	1000	0.199	0.233	0.192

Table 10 shows fixed epsilon effects with observed resulting cost and performance metrics of SVR sorted in ascending order of MAE. Highest R^2 (0.942) with the lowest RMSE (0.063) and lowest MAE (0.048) when target is *Wa*, in kernel type 'Radial' in M1 80:20 and the resulting cost is 100. However, only at a higher resulting cost 1000, the R^2 (0.804) to target *Wd* with a relatively top lowest RMSE (0.108) and relatively top lowest MAE (0.077) in the same kernel type 'Radial' in M1 80:20.

For the kernel type 'Polynomial' in M1 80:20, the R^2 is 0.699 with relatively low RMSE (0.143) and MAE (0.103); when targeting *Wa*; with resulting cost 100. *Wd* for the same model has R^2 0.612, RMSE 0.151 and MAE 0.110; at the same resulting cost 100. This shows that abundance and diversity scores from 'Polynomial' kernel are suggested as unreliable to be a pair of reference from learning the same feature set variables, because the approximately equal error metrics RMSE (± 0.008) and MAE (± 0.007) with a difference in R^2 (± 0.1) as recorded, for *Wa* (0.7) and for *Wd* (0.6). *Wa* is a better target for the kernel based on the R^2. This suggests that the 'Polynomial' kernel struggles to generate scoring for diversity, *Wd*. Thus, this phenomenon establishes that this type of kernel for creating SVR quantifier is not suitable for generating diversity scores.

For the kernel type 'Linear' in M1 80:20, the R^2 is 0.499 with relatively high RMSE (0.172) and MAE (0.119); when targeting *Wd*; with resulting cost 1000. *Wa* for the same model has lower R^2 (0.448), higher RMSE (0.194) and higher MAE (0.164); at resulting cost 100. This is a thematic finding for *Wd* in kernels comparison, that the cost is higher in *Wd* for both 'Radial' and 'Linear' only, but not for 'Polynomial' which has same cost but negligible R^2 with ±0.1 for generating *Wa* and *Wd*, that suggests 'Polynomial' kernel is fundamentally ill-suited for reliably scoring diversity. For visualisation, Table 10 is presented in Figure 8 and 9 showing R^2.

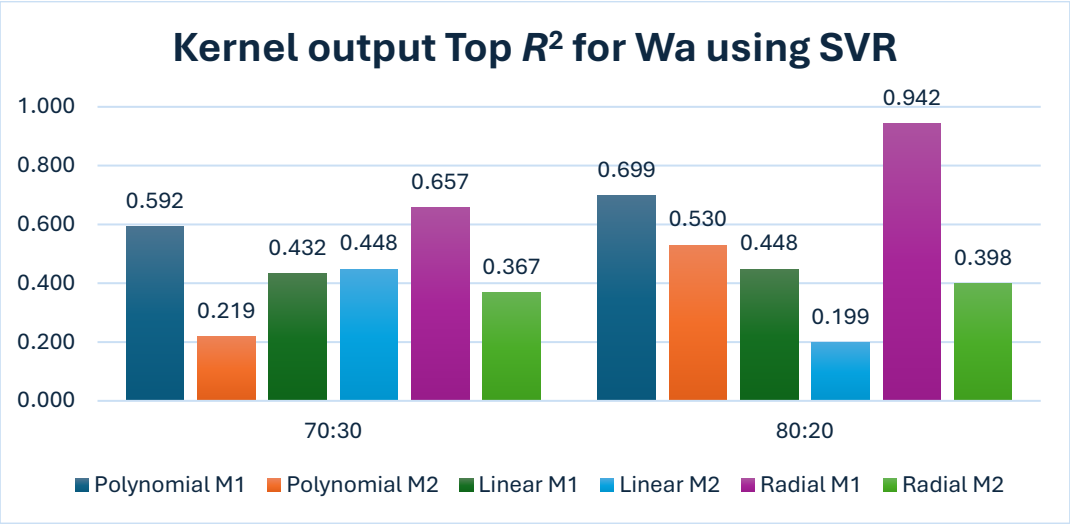


Figure 8. Top R^2 from different kernels for *Wa*

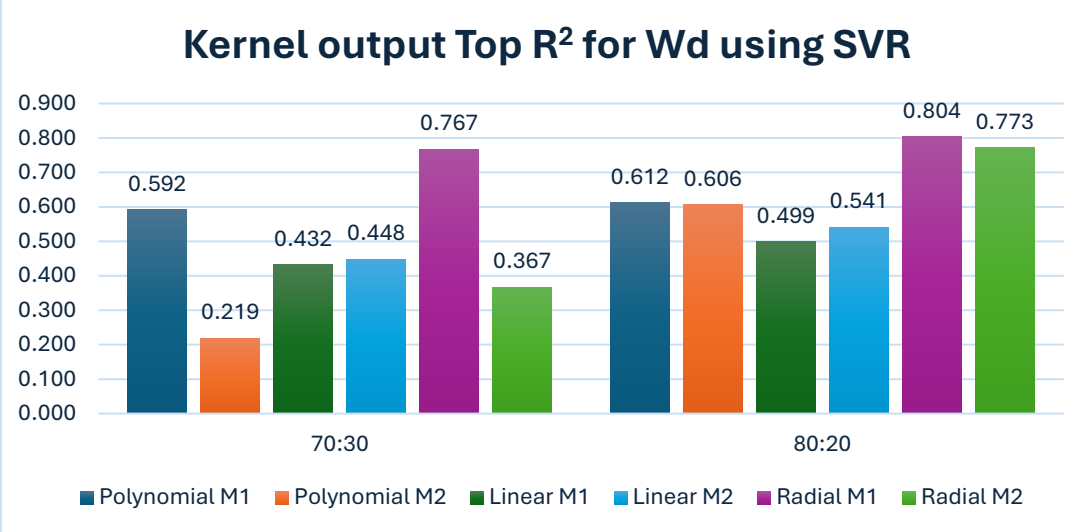


Figure 9. Top R^2 from different kernels for *Wd*

Based on Figure 9, 'Polynomial' M1 80:20 (0.612) has ± 0.020 units R^2 than 'Polynomial' M1 70:30 (0.592). When 'Polynomial' M1 80:20 and 'Polynomial' M2 80:20 (0.606) are compared, their marginal difference ± 0.006, show that the effect of variables that are absent between the two models are not contributing significantly to the performance metrics of the kernel type 'Polynomial', supporting the earlier mentioned unsuitability of 'Polynomial' kernel to be a quantifier for diversity score, *Wd* also because of the approximately equal error metrics incurred at cost 100 for both *Wd* and *Wa*.

'Linear' M1 80:20 and 'Linear' M1 70:30 performed better than 'Polynomial' kernel, with ± 0.067 units R^2 between them, as shown in Table 10. Similarly, this general trend is shown by the suggested best-suited kernel attributing the high R^2 in 'Radial' M1 80:20 (0.942 for *Wa* in Figure 8 and 0.804 for

Wd). Kernel type 'Radial' recorded ± 0.037 units of R^2 that also shows higher cost is incurred for generating *Wd* (cost 1000) than *Wa* (cost 100) based on Table 11.

Table 11 shows the effect of train-test data ratios and number of variables (M1 or M2) in the differences in coefficients of determination, ΔR^2. The compiled values are obtained by subtracting the 70:30 R^2 results from 80:20 R^2 results across all kernels. As mentioned, 'Linear' M1 for both ratios performed better than 'Polynomial' kernel in *Wd.* However, 'Linear' M1 and 'Linear' M2 for *Wa* have different results than the rest of the compared kernels while 'Linear' M1 and 'Linear' M2 show similar trend with the rest. The R^2 for *Wa* found higher in 'Linear' M2 70:30 (0.448, in Figure 8) in 'Linear' M2 80:20 (0.199) contributed to the negative value (-0.250). This shows the effect of ratio in kernel type 'Linear' for the analysis of differences in R^2.

Table 11. ΔR^2 ($R^2_{80:20} - R^2_{70:30}$)

Group	Target	Kernel	ΔR^2
M1	*Wd*	**LINEAR**	0.067
M2	*Wd*	**LINEAR**	0.093
M1	*Wd*	**POLYNOMIAL**	0.021
M2	*Wd*	**POLYNOMIAL**	0.387
M1	*Wd*	**RADIAL**	0.037
M2	*Wd*	**RADIAL**	0.405
M1	*Wa*	**LINEAR**	0.017
M2	*Wa*	**LINEAR**	-0.250
M1	*Wa*	**POLYNOMIAL**	0.107
M2	*Wa*	**POLYNOMIAL**	0.310
M1	*Wa*	**RADIAL**	0.285
M2	*Wa*	**RADIAL**	0.031

Most values are positive meaning the 80:20 training split generally yielded higher than the 70:30 split. Non-linear kernels ('Radial' and 'Polynomial') show difference greater in magnitude than 'Linear' kernel. Analysis of the ΔR^2 indicates that model behaviour varies between feature groups M1 (more variables) and M2 (less variables). For instance, under the 'Radial' kernel, *Wa* generalisation stability shifted noticeably between M1 ($\Delta R^2 = 0.285$) and M2 ($\Delta R^2 = 0.031$), demonstrating that feature dimensionality referring to variables (12 × 5 or 12 × 8) alters kernel responses more than the effect of adjusting train–test proportions. In detail, for concluding the suitable kernel the following paragraphs are presented.

The analysis of ΔR^2 showing the data that are given further improved the SVR model generalisation ability. The negative value ($\Delta R^2 = -0.250$) obtained by the M2 'Linear' kernel is a significant anomaly. It means the 70:30 type showed better results in this specific configuration in M2 that has lesser variables (without geographical information that contains Latitude, Longitude and Elevation) when targeting *Wa*. Thus, this incident in other words, shows that to quantify a location's ability to encourage abundance of bees using its weather properties only, the less data given for training is helpful, but more data given makes for noisy outliers lowering the R^2.

Wd had shown sensitivity to model type based on kernels. The model complexity in 'Linear' kernels show small ΔR^2 gaps (0.067 and 0.093) meaning changing the train–test ratio did not dramatically change the processing behaviour. While using non-linear kernels ('Polynomial' and 'Radial') show a massive divergence between M1 and M2. While ΔR^2 in M1 remains low (0.021 to 0.037), M2 spikes to 0.387 and 0.405 (except 0.031 for *Wa* in 'Radial'). This implies that for *Wd*, the M2 model configuration combined with a complex kernel is extremely sensitive to the amount of training data, showing that

giving it 80% data unlocks a massive performance boost over 70% that it simply cannot achieve with less data. This suggests that the quantifier when run without the three geographical variables, may achieve better performances through the non-linear kernels to generate diversity score; while maintaining that geographical variables may instantly strengthened such inferences.

As mentioned earlier, the 'Radial' kernel processing for *Wa* shows the exact opposite behaviour of *Wd*. M1 'Radial' spikes high (ΔR^2 = 0.285), while M2 'Radial' is showing ΔR^2 at near-zero (0.031). This suggests that for *Wa*, only the M1 model structure is the one that truly utilises the extra 10% of training data when using a 'Radial' kernel.

'Linear' kernels consistently show low ΔR^2 variations across most metrics, except for the negative anomaly on M2 *Wa*. SVR with linear kernel has fewer hyperparameters to adjust, meaning the change in data size does not radically alter its generative capacity. On the other hand, non-linear kernels frequently result in the highest ΔR^2 differences (e.g., 0.405 for M2 *Wd*`Radial and 0.310 for M2 *Wa* Polynomial). 'Linear' kernel approximates a linear function by calculating the inner product within the original feature space, whereas 'Radial' and 'Polynomial' kernels enable nonlinear feature mappings to enhance predictive flexibility (Hitam et al., 2022, Saad et al., 2024). Because non-linear kernels map data into higher-dimensional spaces, they require sufficient data density to find the optimal hyperplane, thus their performance may fluctuate significantly depending on the kernel configuration and available training set size (Izonin et al., 2021, Saad et al., 2024, Jalili et al., 2025).

Therefore, concluding M2 model types paired with non-linear kernels (Radial or Polynomial) for targetting *Wd* benefit the most from an 80:20 split, yielding significantly higher predictive stability than a 70:30 split. For focusing on M1 80:20, based on the Table 11 data which suggests that in targeting of *Wa* and *Wd*, all kernel types are capable of further tuning for different use-case purposes per kernel's strength corresponding to cost parameters allowances (resources capacity) shown in Table 10. However, for picking a choice of kernel in SVR, previous studies (Izonin et al., 2021, Jalili et al., 2025) selected 'Radial' or known commonly as 'Radial Basis Function' (RBF) as the most effective kernel. Thus, the setting of cost 100, epsilon 0.01, and kernel type 'RBF' is suggested for SVR to target *Wa* and *Wd*.

Figures 10 until 15 showed that the deep steep does not occur at 80:20 and 70:30 for all types of machine learning approaches implemented except in Neural Network. This is consistent for two of them, RF and SVR. All the ML models in this study proved an identical general trend. That is, lowering the training dataset size led to reduced model performance as shown in higher training and validation losses. Therefore, the optimum tuning parameters are determined using the train–test split ratio of 80:20 and 70:30, based on literature surveyed in the work of Tan et al. (2021), Vrigazova (2021), and Sivakumar et al. (2024).

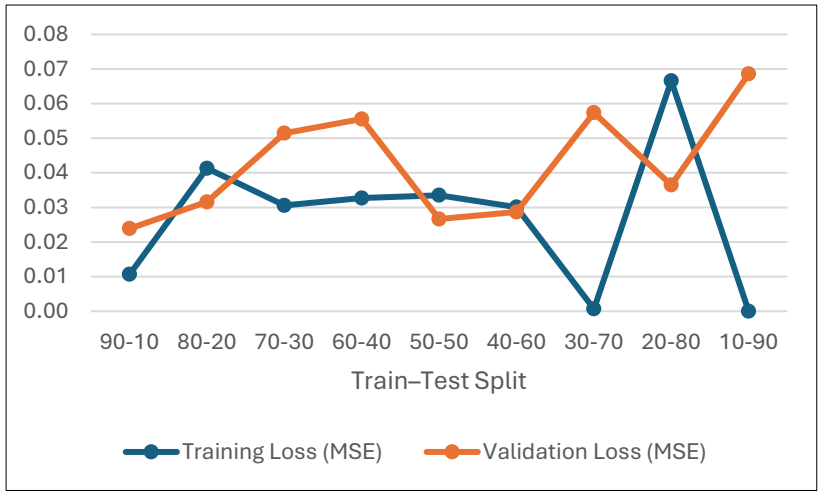


Figure 10. Training Loss and Validation Loss of M1 NN Models for *Wd* across Train–Test Split Ratio

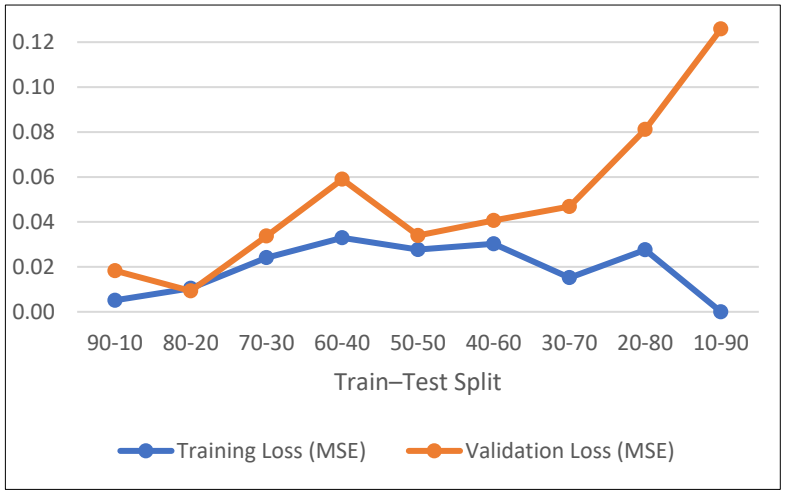


Figure 11 Training Loss and Validation Loss of M1 NN Models for *Wa* across Train–Test Split Ratio

In Figure 10, Neural Network M1 for *Wd*. The effects of training loss are measured in 90% portion of training data to 10% portion training data. The training loss sharper steep is shown from the start at point of 80% portion training data and at 20% portion training data. A relatively low MSE recorded for 90% portion, 30% portion and 10% portion. Meanwhile, Neural Network M1 for *Wa* shown in Figure 11 with the training loss smoother line until MSE recorded for 30% portion and 20% portion of training data. Consistently, for both Figure 10 and Figure 11, the validation loss recorded are often higher than training loss, as shown at point of portions on 70%, 60%, 30% and 10%.

The literature review about *Wd* is limited support on the findings to explain how diversity as a target output variable, in weather score for diversity by Neural Network, is found to be volatile response from portion of testing data from 10% to 90%, in contrast with the generation of *Wa* that is a target output variable in weather score for abundance. However, the observation of diversity takes different harmonics of species while the observation of abundance takes similarities of species to form harmonics. Thus, it is reasoned for Neural Network to have a more difficult time in generating *Wd* than *Wa*. Neural Network plots contained the most fluctuations than Support Vector Regressor and Random Forest as shown in Figures 12 until 15.

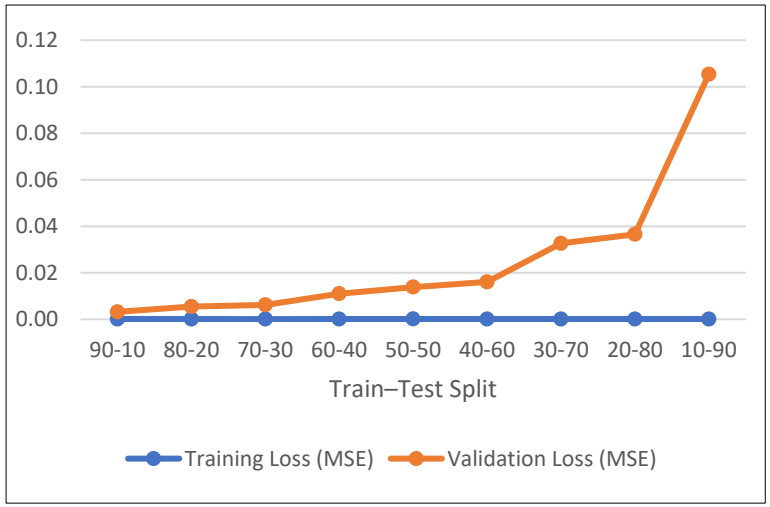


Figure 12. Training Loss and Validation Loss of M1 SVR Models for *Wd* across Train–Test Split Ratio

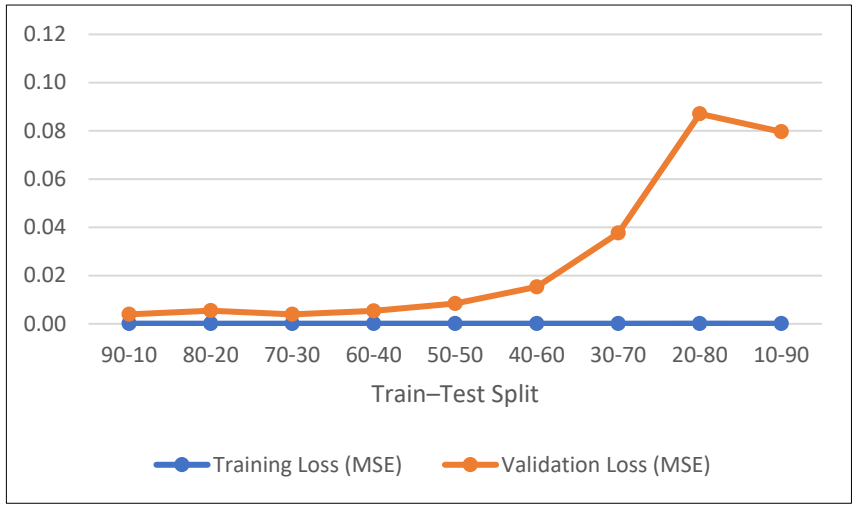


Figure 13 Training Loss and Validation Loss of M1 SVR Models for *Wa* across Train–Test Split Ratio

In Figure 12, Support Vector Regressor M1 for *Wd*. The effects of training loss are measured in 90% of training data to 10% portion training data, which are all visibly close to 0, starting from 0.00010 or 10E-05 (90%) and 8.89E-05 (10%). Both the training loss and validation loss are each a smoother line than in Neural Network, shown from the start at point of 10% of portion training data until 90%. Except for the line plot validation loss, 20% training data was recorded to be 0.037, which is the last point found in the final range of between 0 to 0.04. Following that, 10% training data was recorded 0.105, the point considered as a high validation loss relative to the other points. Meanwhile, the results from *Wa* shown in Figure 13 with the training loss similarly visible close to 0, starting from 8.41E-05 (90%) and 9.10E-05 (10%). Consistently, for both Figure 12 and Figure 13, the validation loss recorded is all higher than training loss, as shown on all the data points.

Similarly with Neural Network, the higher portion (90%) training data in Support Vector Regressor, records larger MSE in training losses than the lowest portion of training data (10%). The validation losses are also recorded in smaller MSE at 90% training data and higher MSE at 10% portion. This shows that the validation losses are higher than training losses in all machine learning approaches when targeting the output variables. The least fluctuations are seen in Random Forest plots, compared to Support Vector Regressor and the most fluctuations in Neural Network.

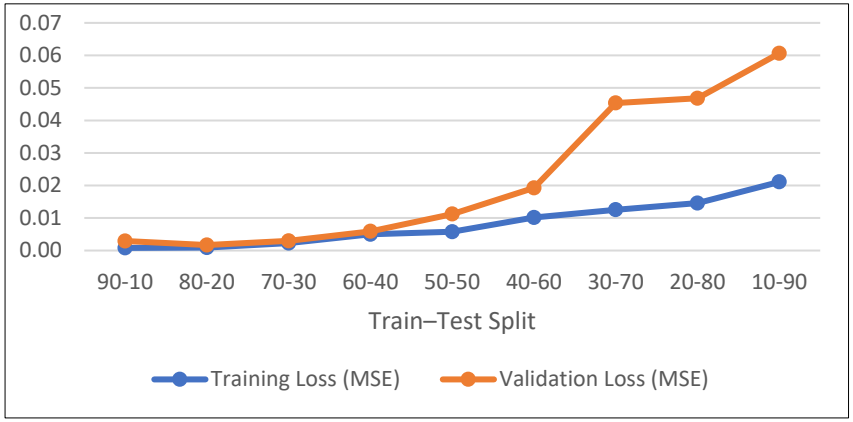


Figure 14. Training Loss and Validation Loss of M1 RF Models for *Wd* across Train–Test Split Ratio

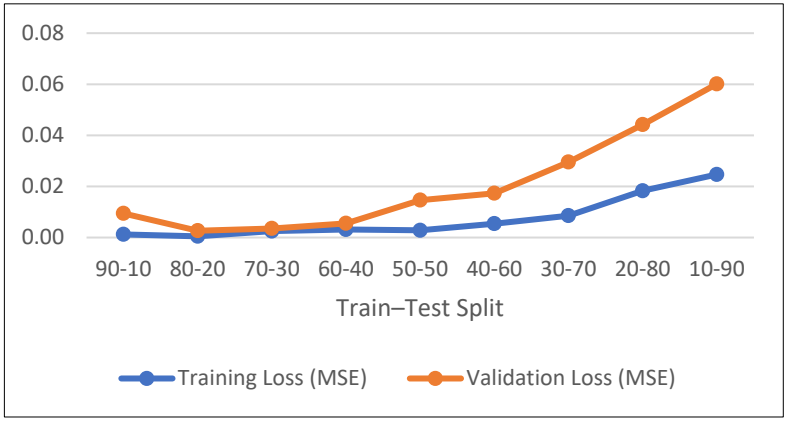


Figure 15. Training Loss and Validation Loss of M1 RF Models for *Wa* across Train–Test Split Ratio

In Figure 14, Random Forest M1 for *Wd*. The effects of training loss are measured in 90% of training data to 10% portion training data, which are all increasing from close to 0, starting from 0.000801 (90%) and 0.0211 (10%). Both the training loss and validation loss are each a relatively smoother line than in Neural Network, shown from the start at point of 10% of portion training data until 90%. Compared to Support Vector Regressor plots, the training loss and validation loss records of MSE in Random Forest are both showing visible pattern of relatively steady increase from datapoint to datapoint.

Meanwhile, the results from *Wa* shown in Figure 15 with the training loss showing similar pattern with *Wd*, starting from 0.00124 (90%) and 0.0247 (10%). Consistently, for both Figure 14 and Figure 15, the validation loss recorded is all higher than training loss, as shown on all the data points. The convergence points on approximately equal training loss and validation loss occurred at train–test split ratio 80:20, 70-20 and 60-40.

Similarly with Neural Network and Support Vector Regressor, the higher portion (90%) training data in Random Forest as a dataset, has larger MSE of training losses than the lowest portion of training data (10%). However, the validation losses are recorded in smaller MSE at 90% training data and higher MSE at 10% portion. This shows that the validation losses are higher than training losses in all machine learning approaches when targeting the output variables. Less fluctuations are seen in Support Vector Regressor plots than in Neural Network.

Considering the 70:30 and 80:20 train–test split and effects from combination of features (M1 or M2) with tuning parameters for RF, NN and SVR, the findings conclude for the eight quantifiers. Through one of the approaches, two selections are made from the following listed:

(a) M1 Abundance 70:30
(b) M1 Abundance 80:20
(c) M1 Diversity 70:30
(d) M1 Diversity 80:20
(e) M2 Abundance 70:30
(f) M2 Abundance 80:20
(g) M2 Diversity 70:30
(h) M2 Diversity 80:20

This study selected models (b) and (d). The underlined values in Table 12 show the smallest training loss and validation loss are found in SVR and RF respectively. Based on the performance of ML models in training and loss validation of MSE, the most optimum tuning parameters are NN model with three-layer neurons (15, 15, 15), 'Adam' optimiser, and threshold value of '0.5'. For SVR model, parameters tuned at cost '100', epsilon '0.01', and 'RBF' kernel. For RF model parameters are tuned at Node Size '3' and '10' trees.

Table 12. Training and Validation Loss of the Most Optimum RF, SVR and NN Models

Group	Training Loss (MSE)	Validation Loss (MSE)	Target	ML
M1	0.000898	<u>0.001686</u>	*Wd*	RF
M2	0.013181	0.024811	*Wd*	RF
M1	0.000479	0.002662	*Wa*	RF
M2	0.011630	0.026686	*Wa*	RF
M1	0.000110	0.005482	*Wd*	SVR
M2	0.019700	0.044255	*Wd*	SVR
M1	<u>0.000084</u>	0.005451	*Wa*	SVR
M2	0.019989	0.046370	*Wa*	SVR
M1	0.017147	0.020310	*Wd*	NN
M2	0.039617	0.039931	*Wd*	NN
M1	0.009270	0.013137	*Wa*	NN
M2	0.047946	0.043984	*Wa*	NN

RF is chosen due to its combined decision tree structure, which leverages multiple decision trees to introduce greater randomness and model diversity, as found in this study, agreeing with similar applications of random forest that utilise different input data combinations (Sakaa et al., 2022; Rhodes et al., 2023). This selection is particularly advantageous for small datasets, where limited training samples may restrict the model's ability to capture unseen data relationships, as noted by Sivakumar et al. (2024). This suggests that RF provides a means of achieving balance between underfitting and overfitting while being best suited for small dataset applications, such as in limited abundance data in studying specific traits of fish (Luan et al., 2020) and in limited data when studying biomedical vaccine efficacy trials (Han et al., 2021). Therefore, this study resorted to selecting RF to quantify the weather score for abundance and diversity, with the M1 model of 80:20 split, $NS = 3$ and '10' trees.

Limitation and recommendations:

The study is limited by one-choice to use only one type of grouped variables, M1 or M2 for generating *Wd* and *Wa*. The concluded choice has its own strengths and limitations with the main feature difference being the inclusion of geographical variables, compared to M2 (without). Thus, a methodological limitation of this study is the dependence on numerical geospatial coordinates (latitude, longitude, and elevation) rather than visual raster or land-cover imagery. While spatial imagery can visually verify ground conditions such as deforestation or vegetation density, the numerical coordinates and elevation data act as direct alternatives for measuring local topography potentials in line with bipartite network approach used in the original study that is based of Bipartite Network Modelling Framework (Liew et al. 2023). Consequently, the machine learning models tested become part of the library of quantification techniques documented that work on the quantification potentials of the location rather than its visual representation. Future work could integrate raster-based vegetation indices to cross-verify these spatial anchors against real-time ecological degradation and consider input doubling method for small dataset using SVR with RBF kernel as shown in work of Izonin et al. (2021).

CONCLUSION

The methodology for quantification of the weather variables is decided after comparisons of machine learning models created from Random Forest (RF), Neural Network (NN) and Support Vector Regressor (SVR) in three sequential steps: (1) Grouped weather variables executed in models of 5-variable or 8-variable, in 70:30 and 80:20 train–test split, (2) Measured performance metrics for each model using coefficient of determination (R^2) and mean absolute error (MAE) and (3) Measured training loss and validation loss for each model using mean square error (MSE).

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